Classroom Energy Control Using Occupancy and Daylight

An independent LTspice simulation study
October 2026

Abstract. I modelled classroom lighting and an interactive whiteboard using my school's timetable and equipment ratings. My main comparison assumes manual lighting is already off for half of the 08:00-17:30 window. Under one fixed daylight profile, perfect detection and a three-minute delay, estimated savings across 24 classrooms are 21.4% of lighting-plus-board energy. I separately tested synthetic detection gaps in LTspice. The energy totals are LTspice outputs; school-scale projections are conditional estimates, not measured school savings.

1. Question and model

I investigated how occupancy and daylight control affect energy use and required service. My school has 24 classrooms with four fluorescent fixtures and one interactive whiteboard each. I modelled two schedule types: 12 regular-use and 12 extended-use classrooms.

Study scope. I focused on lighting and the interactive whiteboard because occupancy and daylight control can change their operating times. Our classrooms have no separate computers. I excluded heating, cooling and other loads; heating would require a separate temperature and comfort model.

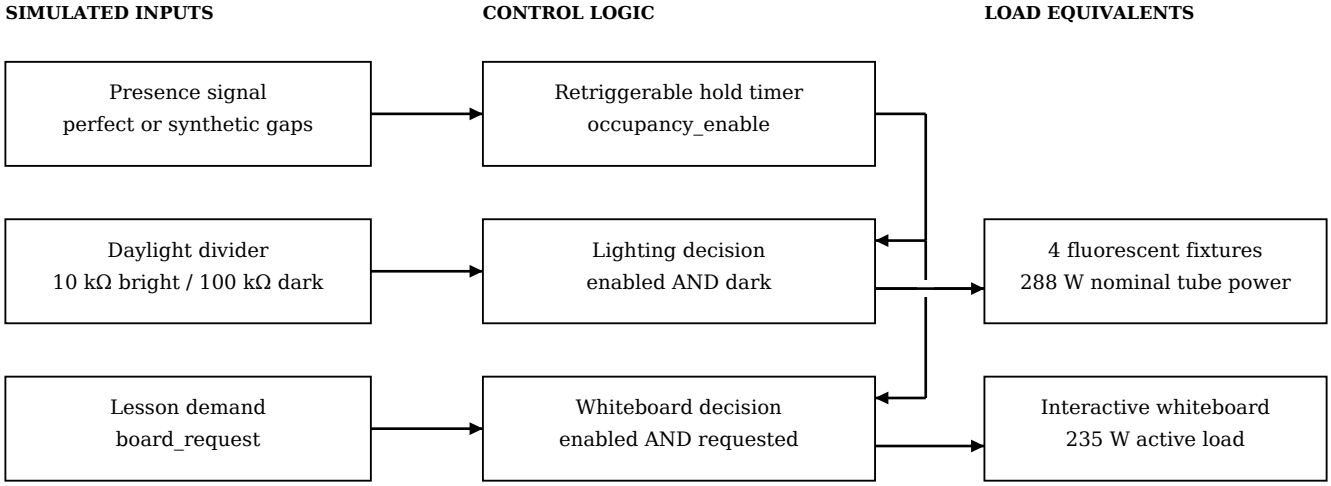


Figure 1. Functional overview of my simulated controller. I used actual occupancy as a separate reference to score errors; the controller responds to the sensor signal.

I used the nominal 4 × 18 W tube rating for each fixture [1] and the whiteboard's published 235 W active rating [2]. At 230 V RMS and 50 Hz, I represented each fixture by 734.7 Ω and the board by 225.10 Ω. My 5 V divider produces about 4.12/1.60 V for the chosen bright/dark resistances, with a 2.5 V decision threshold.

Modelled equipment	Quantity	Nominal power
Fluorescent fixtures	4 per classroom	288 W total, excluding ballast losses
Interactive whiteboard	1 per classroom	235 W while requested

2. Scenarios and energy results

At my school, regular lessons run from 08:00 to 16:05, with 370 minutes of teaching. Half the classrooms have an additional 16:15-17:00 session. Around 4-5 students remain during breaks, so I kept occupancy active throughout breaks and the transition to the extra session. I assumed final departures at 16:05 or 17:00 and set board demand to follow lessons, with the board off during breaks.

I fixed one main baseline: manual lighting is on for 4 h 45 min and off for 4 h 45 min within 08:00-17:30. This already includes manual energy-saving behaviour. I used the schedule below for both classroom types. It covers every occupied period with insufficient daylight; no saving is credited to leaving occupants in darkness.

Time	Natural daylight	Manual lighting
08:00-09:00	Insufficient	On
09:00-13:45	Sufficient	Off
13:45-15:00	Sufficient	On
15:00-17:30	Insufficient	On

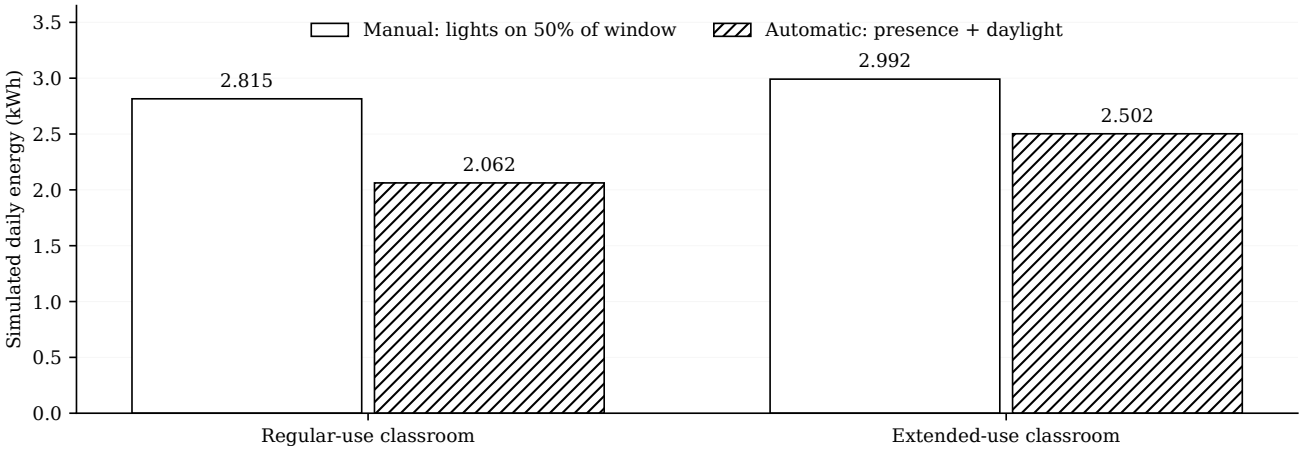


Figure 2. LTspice results for lighting plus whiteboard energy.

Classroom type	Manual / automatic	Reduction
Regular use	2.815 / 2.062 kWh	26.75%
Extended use	2.992 / 2.502 kWh	16.36%

I held board demand identical in both cases, so savings come from lighting. Automatic control uses perfect detection and a reference 180 s delay. Percentages use the combined lighting and board energy as the baseline.

Result check. All four main runs reported zero unwanted lighting and board off-time. Vacant-room energy was 407.74/143.91 Wh for manual regular/extended use and approximately 14.39 Wh for each automatic case.

3. Delay versus unwanted interruptions

To test missed detections, I ran a separate continuously dark experiment over 08:00-17:30 with delays of 0, 60, 180, 300 and 600 s. I inserted occupied-room detection gaps of 20 s at 08:55, 120 s at 09:20, 240 s at 12:25 and 420 s at 14:00.

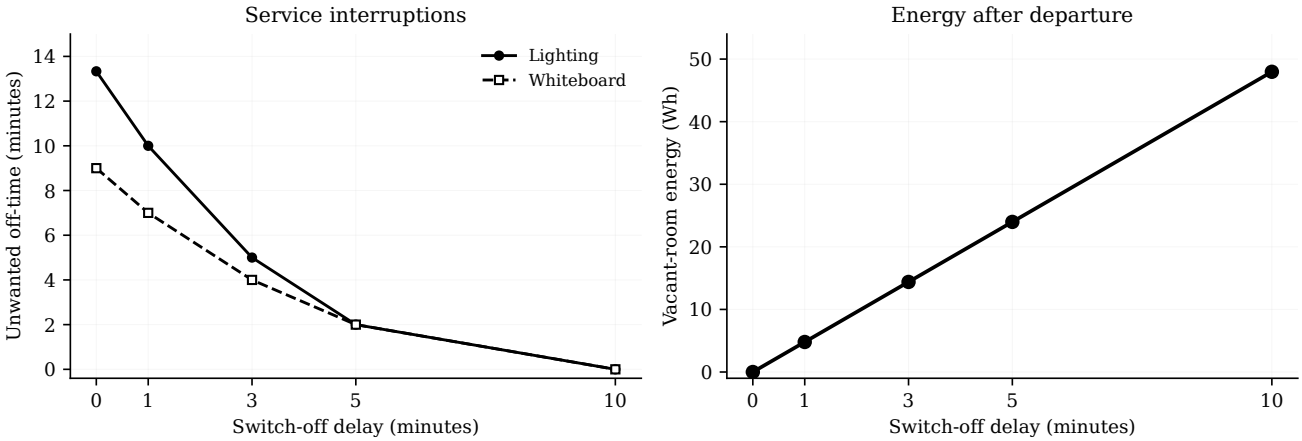


Figure 3. Unwanted off-time and vacant-room energy in the synthetic-gap experiment. Both classroom schedules have the same interruption durations; vacant energy is approximately equal at the plotted precision. The board is only required during lessons.

Delay	Lighting off	Board off	Vacant energy
0 s	800 s	540 s	≈0 Wh
60 s	600 s	420 s	4.80 Wh
180 s	300 s	240 s	14.39 Wh
300 s	120 s	120 s	23.98 Wh
600 s	0 s	0 s	47.97 Wh

I found that longer delays protected against missed detections while leaving the lights on longer after departure. A 300 s delay still allowed a 120 s interruption during my longest, 420 s gap. The 600 s setting covered all four test gaps.

I also found that lower energy could indicate failed service. At 180 s, the regular-use stress model consumed 3.750 kWh while reporting 300 s of lighting interruption and 240 s of board interruption. At nominal powers, these outages represent approximately 39.64 Wh of unmet demand (288 × 300/3600 + 235 × 240/3600). I therefore did not count this avoided consumption as successful saving.

Measurement and interpretation

I calculated load energy by integrating source-delivered power and counted vacant-room energy only when actual occupancy was zero. I counted unwanted off-time only when a load was required: lighting needed occupancy and darkness; the board needed occupancy and lesson demand. I assessed energy alongside interruption duration to avoid mistaking failed service for saving. Dikel and Newsham [4] discuss this delay/reliability trade-off.

4. School-scale impact and cost

To estimate the potential impact at my school, I scaled the selected loads to 24 classrooms: 12 regular-use and 12 extended-use rooms. I used the changing-daylight scenario with perfect detection and a 180 s delay.

Automatic lighting is on 08:00-09:00 and from 15:00 until 16:08 or 17:03. Compared with the manual schedule in Section 2, it avoids unnecessary early-afternoon lighting and shortens lighting after departure.

Aggregation and operating days

I calculated daily saving as 12 × (regular manual − regular automatic) + 12 × (extended manual − extended automatic). I assumed 20 school days per month and 180 per year, repeating the same scenario each day. I excluded weekends and holidays. This annual extrapolation does not account for seasonal changes.

Period	Manual (kWh)	Automatic (kWh)	Saved (kWh)
One school day	69.68	54.77	14.91
Month: 20 days	1,393.65	1,095.47	298.18
Year: 180 days	12,542.86	9,859.26	2,683.60

Estimated impact: approximately 14.9 kWh per school day, 298 kWh per 20-day month and 2.68 MWh per 180-day school year. The reduction is about 21.4% of the modelled lighting-plus-board baseline.

Financial translation

I estimated avoided electricity cost as saved kWh × applicable variable price (TRY/kWh). I used illustrative prices of 3, 5 and 7 TRY/kWh; the applicable school tariff would determine actual costs [5]. I held each price constant across the modelled days and did not assume that fixed or demand charges would decrease.

Illustrative price	Monthly saving (TRY)	Annual saving (TRY)
3.00 TRY/kWh	895	8,051
5.00 TRY/kWh	1,491	13,418
7.00 TRY/kWh	2,087	18,785

Using the illustrative 5.00 TRY/kWh input, I estimated avoided expenditure of approximately 75 TRY per school day, 1,491 TRY per modelled month and 13,418 TRY per modelled school year. These are gross avoided electricity costs, before equipment, installation and maintenance expenses.

Interpreting the percentage

The 50% assumption describes manual lighting's on-time, not the percentage of energy saved. The estimated 21.4% reduction applies to the selected lighting-plus-board loads under this fixed schedule.

5. Timing, verification and limits

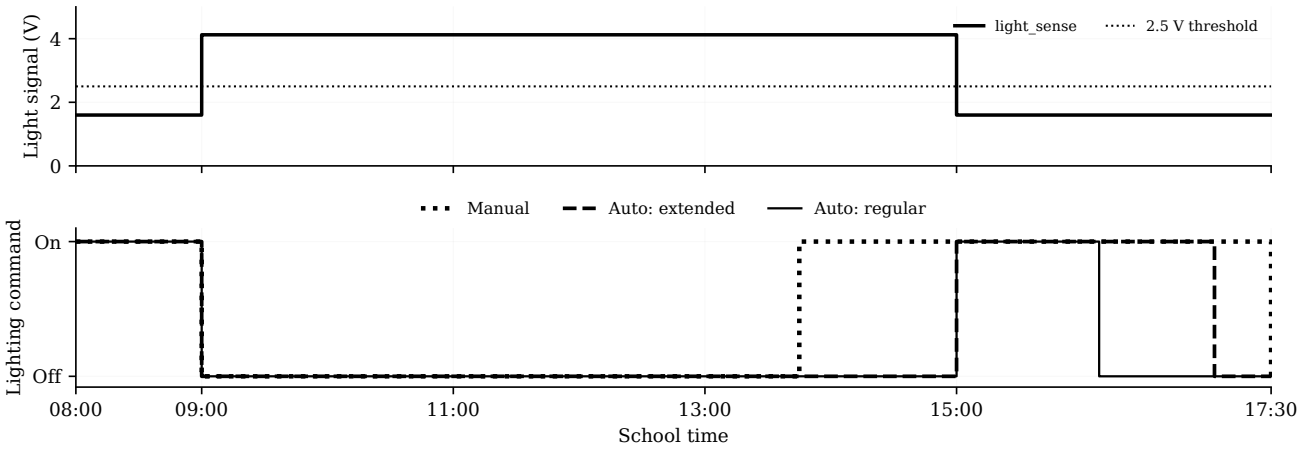


Figure 4. Reconstructed main-case timing. The dotted line shows my fixed manual schedule; the other lines show automatic commands reconstructed from the model schedules. Manual lighting turns on at 13:45; automatic lighting waits until 15:00.

I used TIME_SCALE=60: one simulation second represents one real minute. My 08:00-17:30 window corresponds to 34,200 real seconds, or 570 simulation seconds. I scaled schedules, timers and power integrals consistently, kept AC at 50 Hz, and limited time steps to 200 µs. Energy in Wh is the scaled integral of -V(power) × I(V1), divided by 3600 [3].

10-minute verification	**Real-time run**	**Compressed run**
Simulation duration	600 s	10 s
Load energy	57.946387 Wh	57.946387 Wh
Vacant-room energy	14.390833 Wh	14.390833 Wh

Limits and conclusion

The model uses rated-power resistive equivalents and synthetic daylight and detection gaps. I did not measure illuminance, device consumption or field savings. Ballast losses, standby and controller energy, nonlinear load behaviour, switching wear and board restarts are excluded. The tested delays establish performance for the chosen gaps rather than a universal optimum. I used LTspice 26.1.1 for Windows and retained the results as screenshot transcriptions.

My results suggest that daylight control and switching off after departure can reduce lighting consumption under the tested assumptions. I conclude that energy use and required service must be evaluated together. Before recommending an installation, I would need field measurements of equipment consumption, detection performance and manual switching behaviour.

References

[1] GESTAS. GFST-T8RS T8 luminaire catalogue, p. 57, 4×18 W option. Manufacturer catalogue.

[2] Vestel. 65-inch IFE interactive whiteboard, active consumption 235 W. Product and specifications.

[3] Analog Devices. LTspice Simulation Commands Reference. Sources [1]-[3] accessed 15 October 2026.

[4] Dikel, E. E. and Newsham, G. R. (2014). Unlocking the potential energy savings from shorter time delay occupancy sensors. LD+A, 44(12), 54-56. NRC manuscript.

[5] EPDK. Electricity bill calculation and tariff applicability. Accessed 15 October 2026.